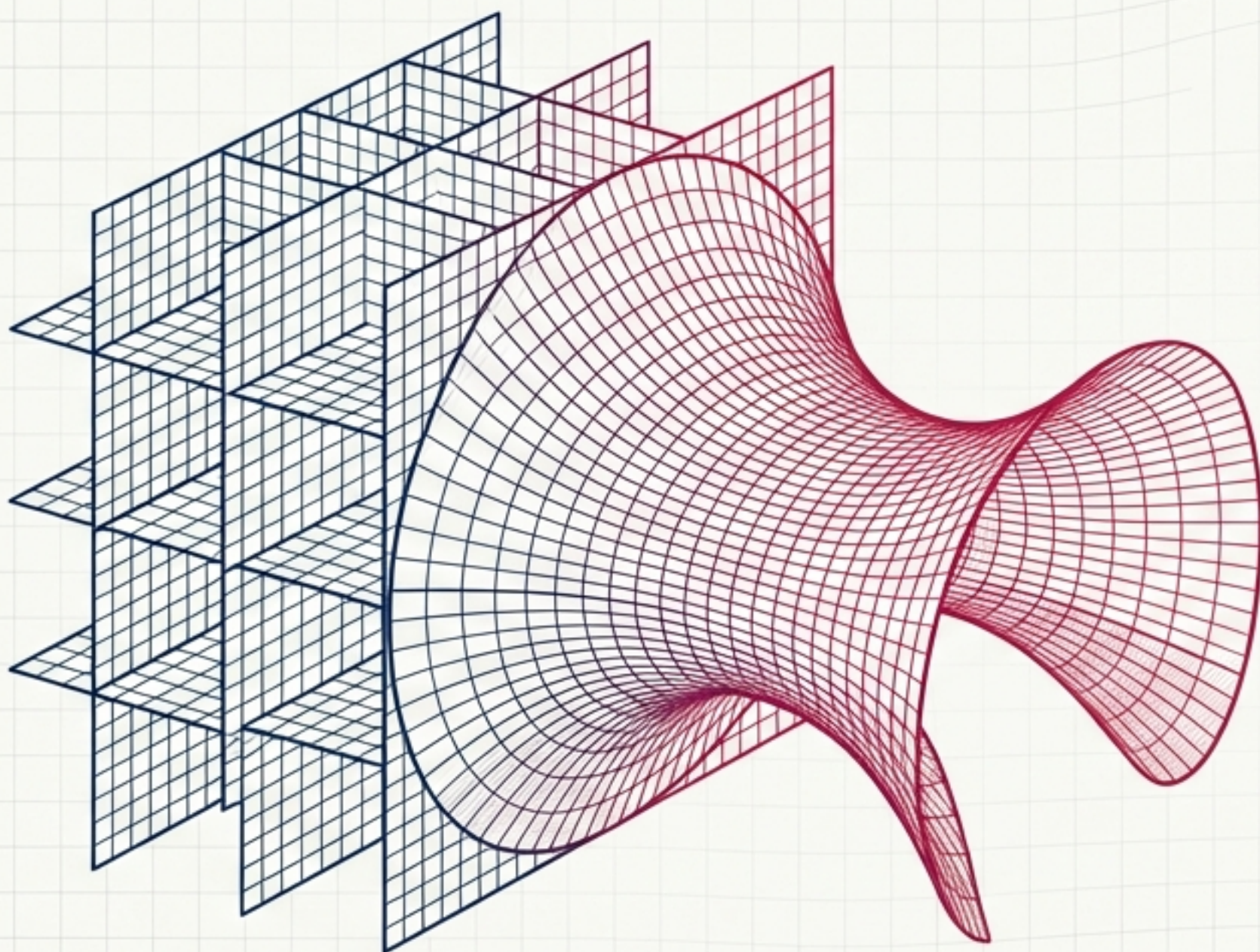




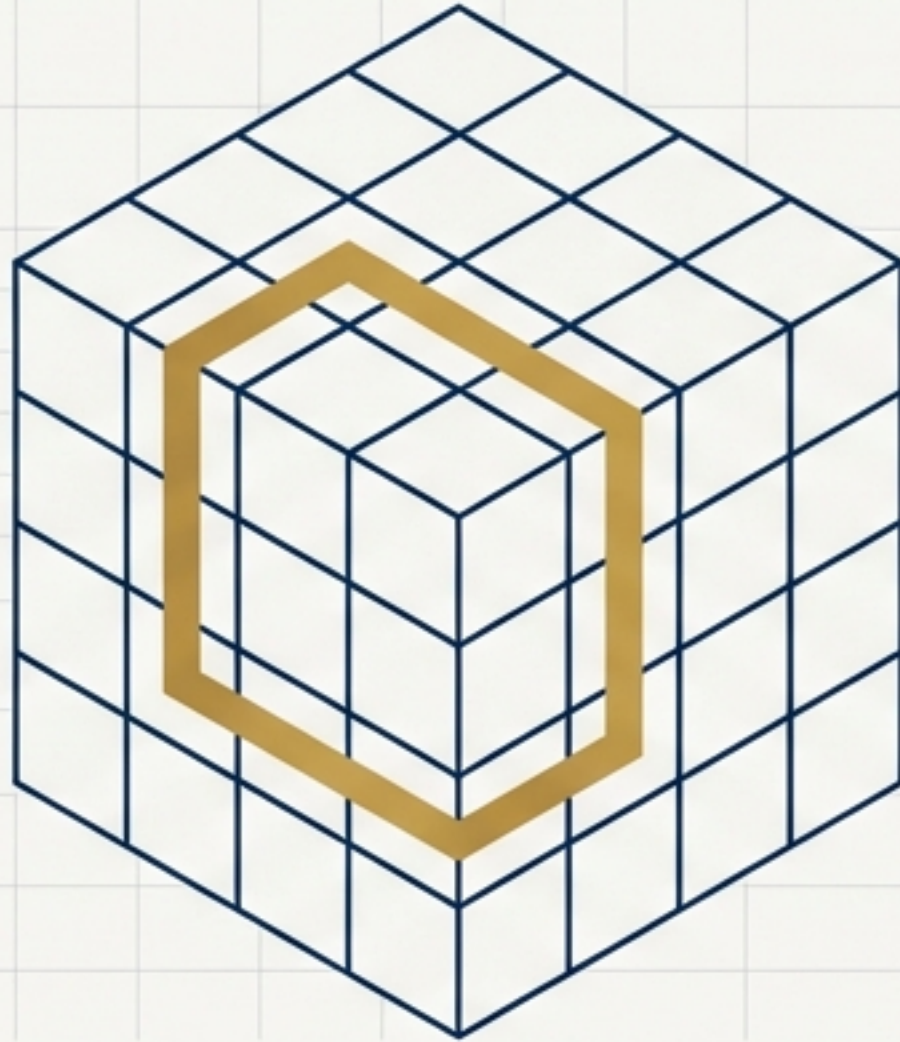
Topological Silicon

The SHD-CCP Architecture

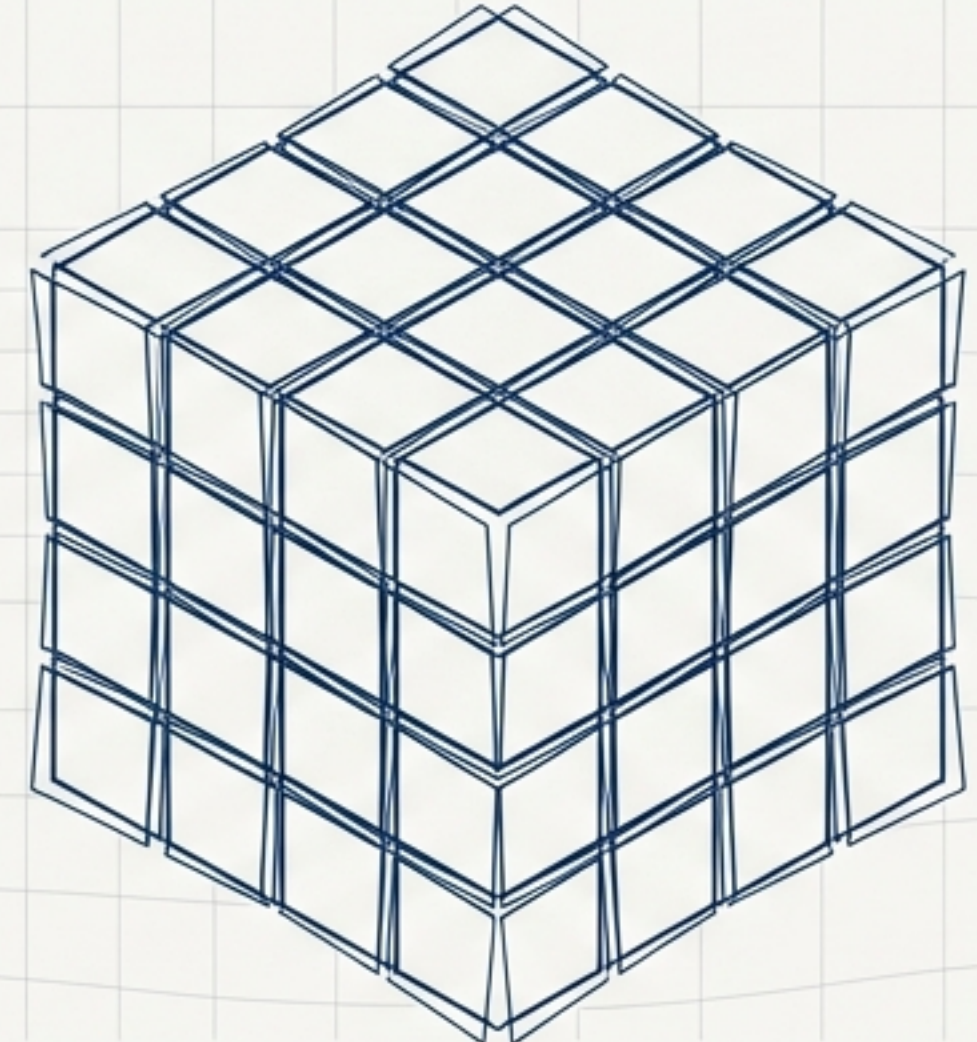
Escaping Euclidean Limits via Continuous
Cyclic Projection and Spinor Stabilization

BioChain AI Architecture Research

The Physical Constraint of Continuous Generation



Exact Integer Detection via TQF



Infinite Expansion = State Drift

The Baseline

The **4x4x4 framework** utilizes Rational Trigonometry and the Triple Quad Formula (TQF).

Result: Exact, integer-based redundancy detection.

The Bottleneck

Fibonacci scaling and Fokker-Planck evolution make tracking **absolute spatial coordinates** computationally prohibitive.

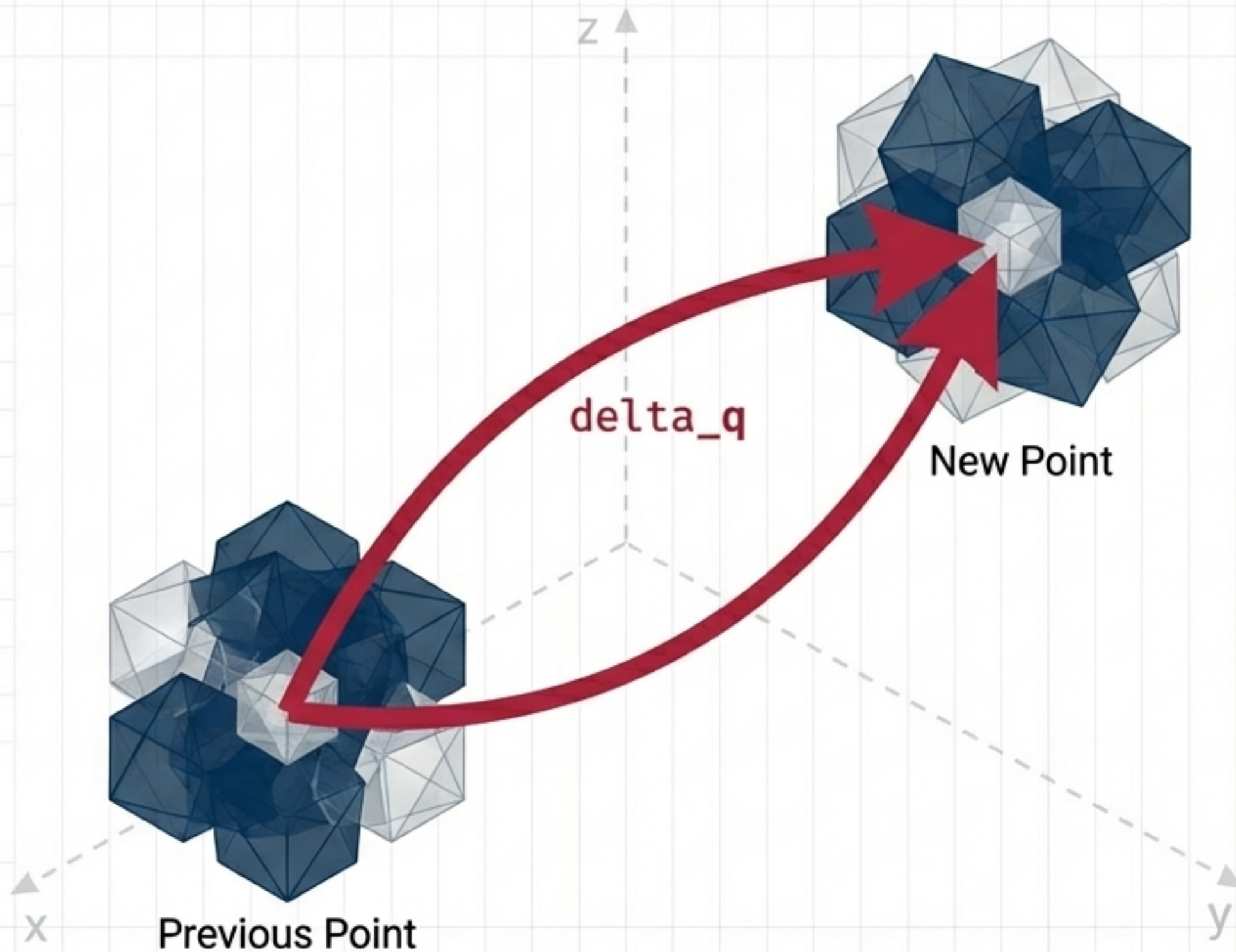
The Threat

Infinitesimal floating-point errors accumulate, eventually blurring the Prime Skeleton geometric target.

From Affine Flaws to Topological Silicon

Parameter	Legacy Affine Framework	Topological SHD-CCP Protocol
State Tracking	Absolute Grid Coordinates (x,y,z)	Quaternionic Delta Encoding
Expansion Constraint	Orthogonal Bifurcation (Infinite Volume)	Hyperbolic Projection (Bounded Volume)
Memory Cost	Linear Path Logging	Trefoil Symmetry Folding (O(1) Lookup)
Drift Resolution	Floating Point Accumulation	Spinor Cyclic Reset (Automated Purge)

Escaping Physical Coordinates: Delta Encoding



Core Directive

We no longer store the universe. We exclusively track the exact rotational sequence required to move from the previous point.

$$\text{State}_{\text{new}} = \text{delta_q} * \text{State}_{\text{old}}$$

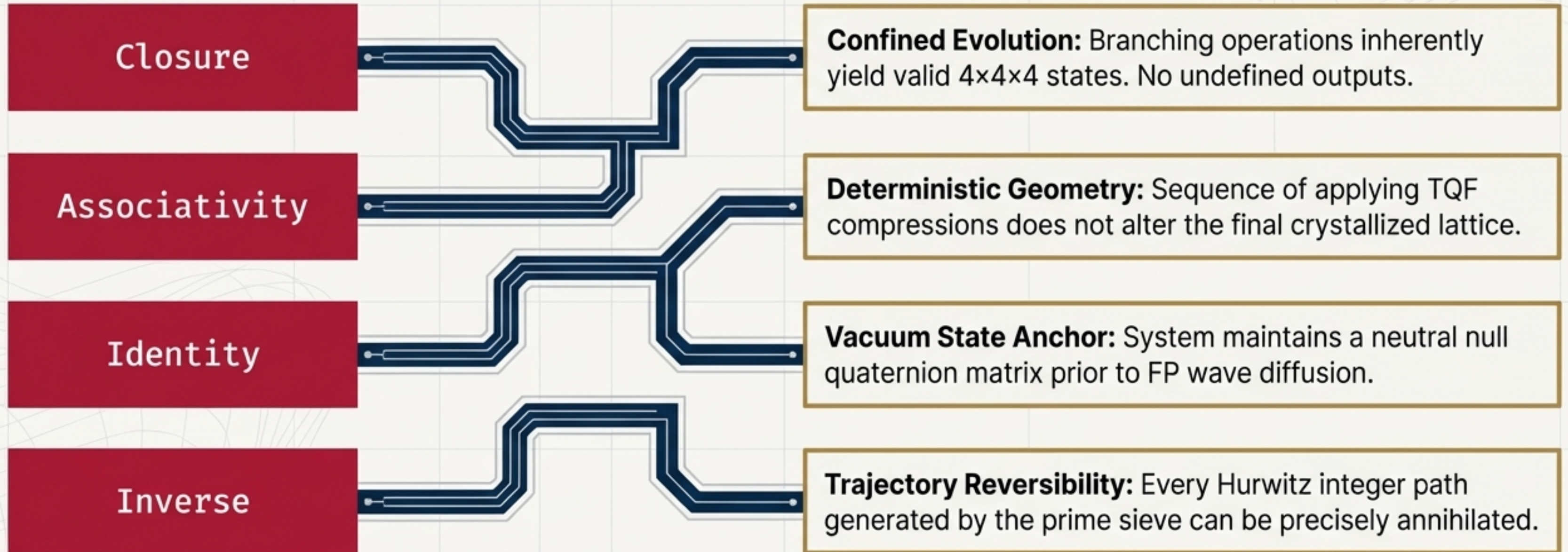
$$\mathbf{q} = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Hardware Impact

Eliminates absolute coordinate overflow by turning geometric expansion into a purely procedural chain of relative differentials.

Translating Algebraic Axioms to RTL Constraints

The 4 fundamental axioms of a Mathematical Group strictly govern transitions within the $4 \times 4 \times 4$ lattice tensor.



Bounding the Infinite: $SL(2, \mathbb{C})$ Dynamics

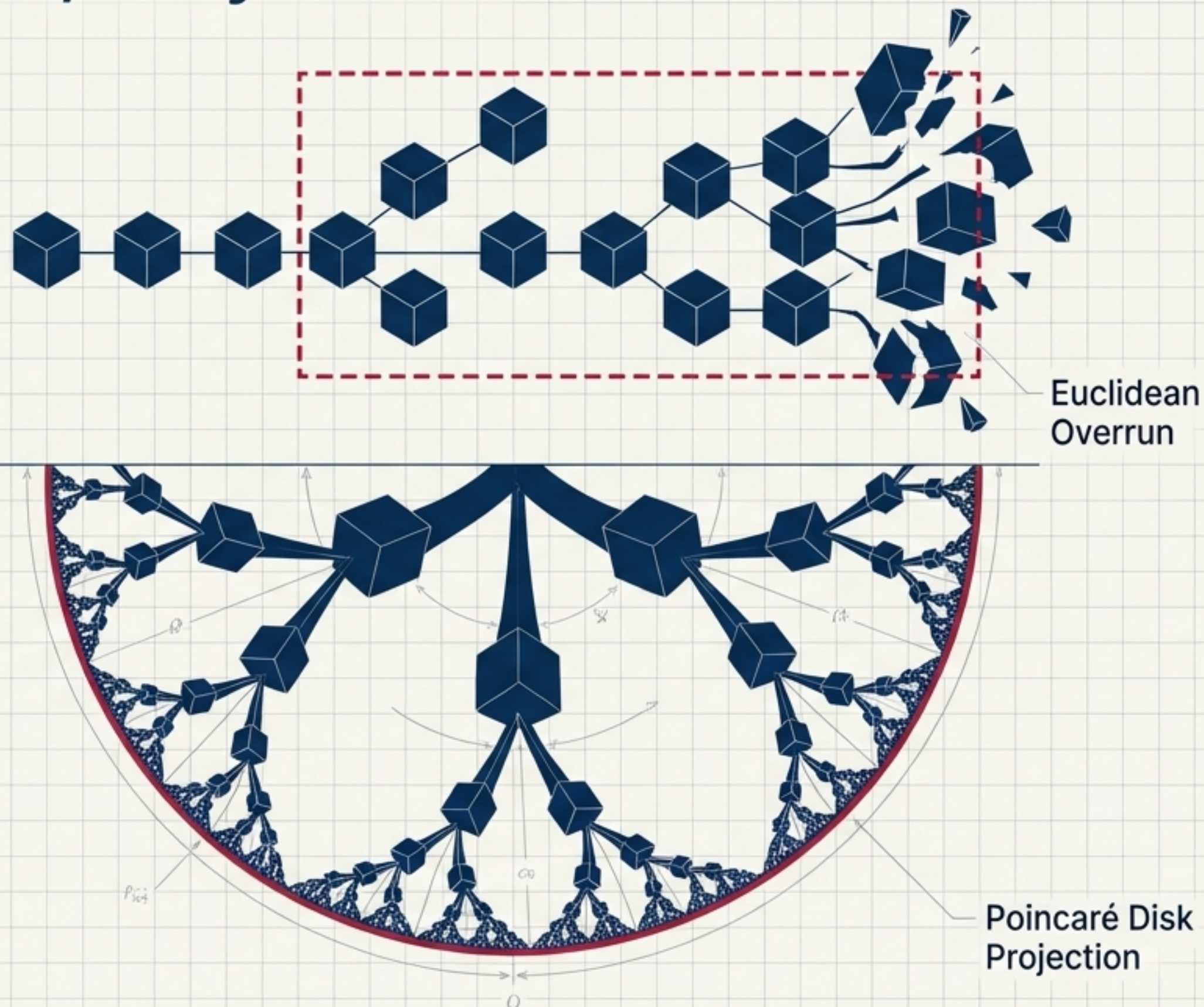
The Mechanism

Branches are projected from standard $SO(3)$ rotation into negatively curved hyperbolic manifolds using $SL(2, \mathbb{C})$.

$$T_{\text{hyperbolic}} = \mathbf{H} * T_{\text{euclidean}}$$

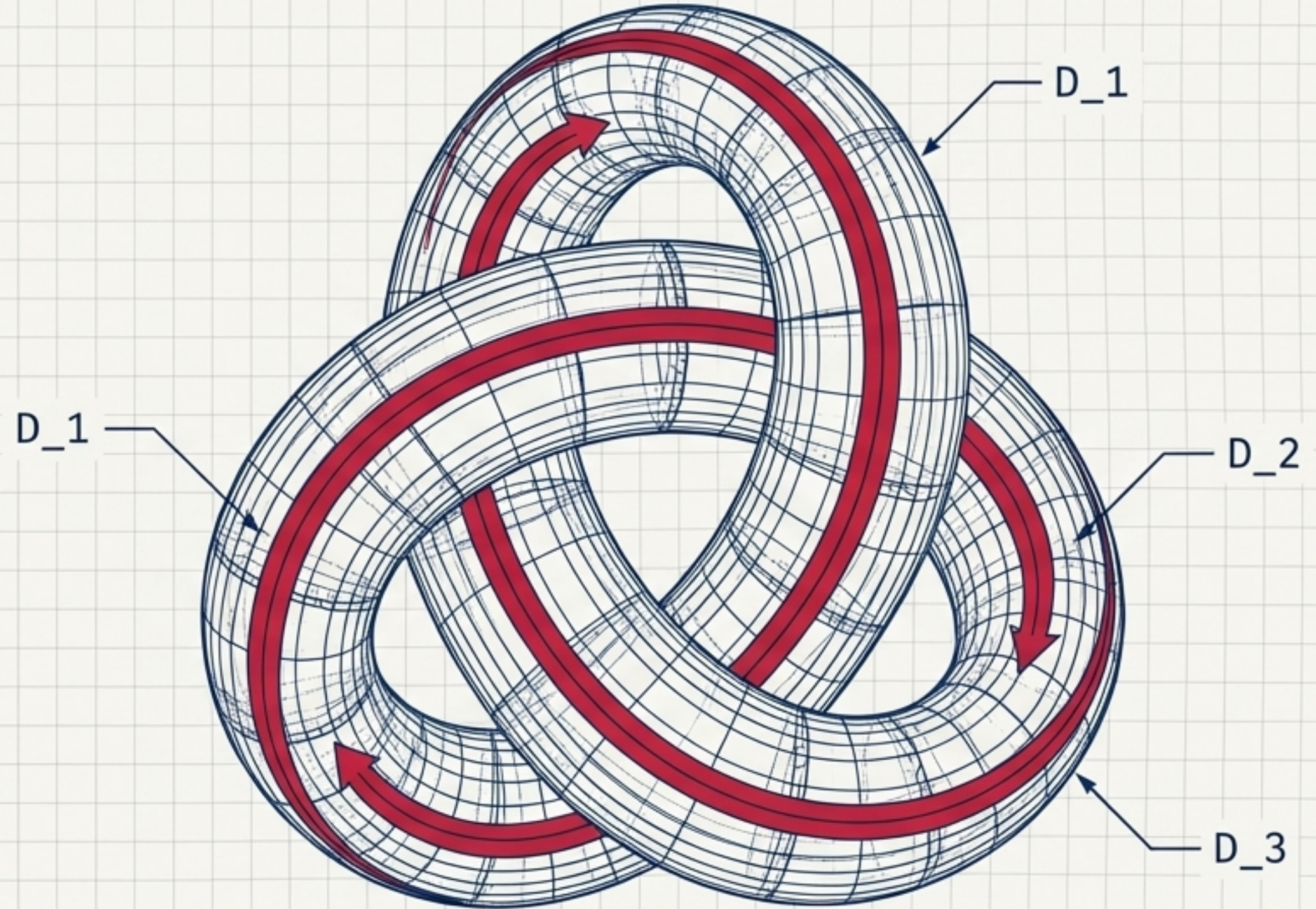
The Hardware Takeaway

Creates mathematically infinite branching inside a finite FPGA volume. Boundary conditions are heavily optimized by pre-calculating fixed projection matrices ($\mathbf{H}$).



The Rule of 3, Topologically Formalized

The linear 3-cycle trajectory is upgraded. The three spatial data streams are constrained to a Trefoil manifold, wrapping the torus exactly 3 times in one direction and 2 in another.



Parametric Compression Constraints

$$x(t) = \sin(t) + 2\sin(2t)$$

$$y(t) = \cos(t) - 2\cos(2t)$$

$$z(t) = -\sin(3t)$$

D_3 Dihedral Symmetry and O(1) Lookup

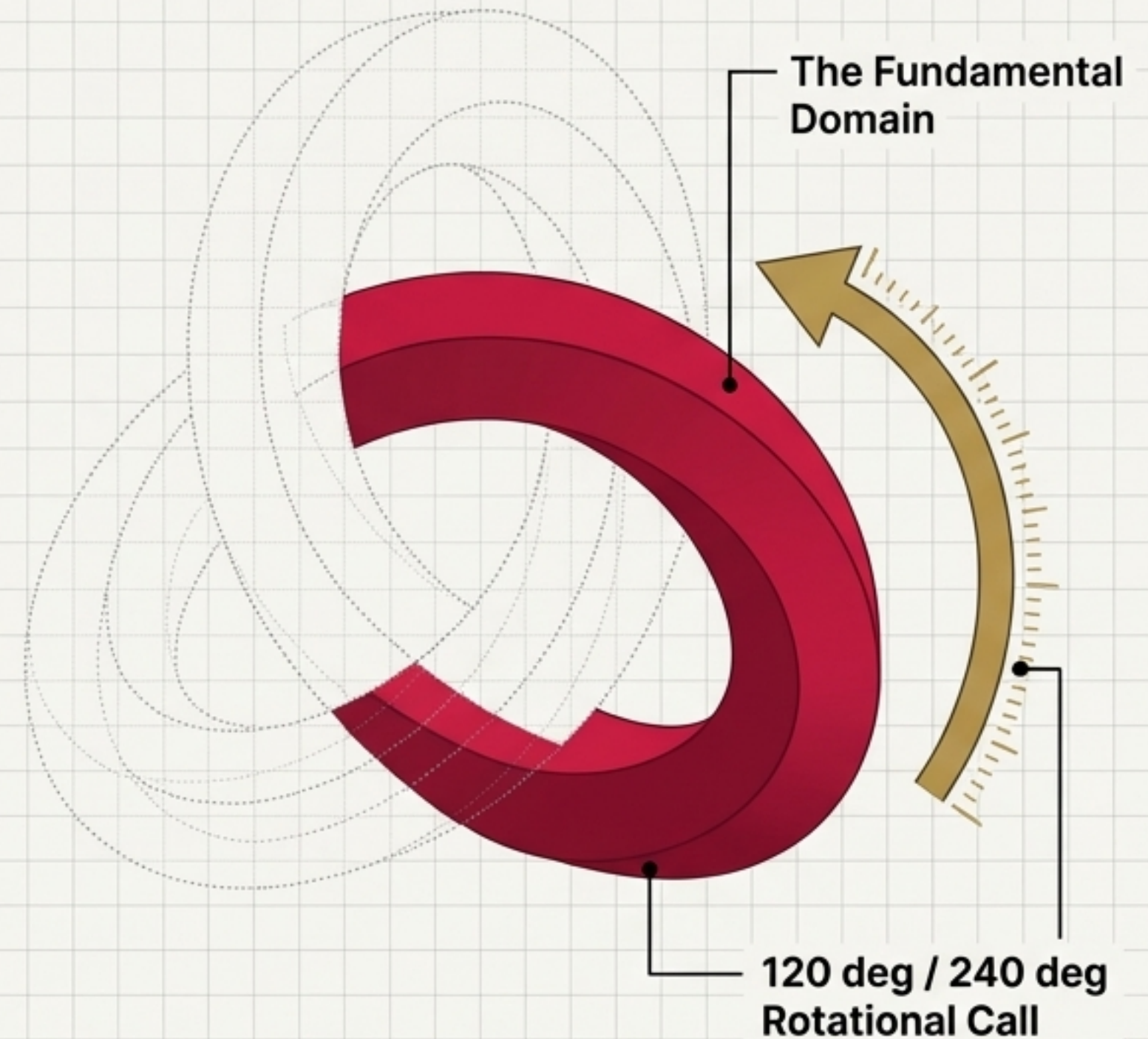
The Hardware Hack

The Trefoil possesses 3-fold rotational symmetry. We only store the Fundamental Domain (the non-repeating 1/3 segment) in memory.

The Retrieval Mechanic

When equations demand out-of-domain coordinates, the protocol retrieves the fundamental domain and applies a static rotational matrix.

66.6% BRAM Reduction via deterministic, constant-time O(1) lookup.



The Vulnerability of the Prime Skeleton

1. The Engine

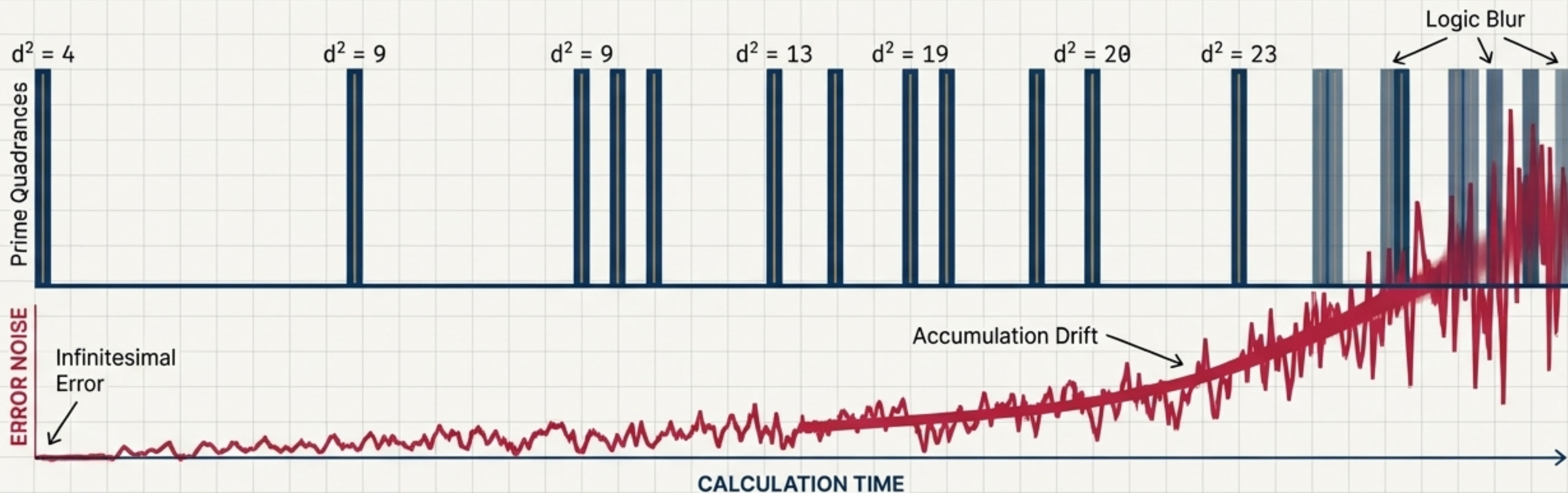
The continuous Fokker-Planck equation acts as a prime number sieve. Stable probability modes map to prime quadrances (d^2), bypassing irrational numbers.

2. The Flaw

Continuous FP simulations naturally accumulate infinitesimal floating-point errors over time.

3. The Threat

Left unchecked, this fractional drift eventually blurs the precise logic of the target geometry.



The Spinor Cyclic Constraint

Quantum Borrowing

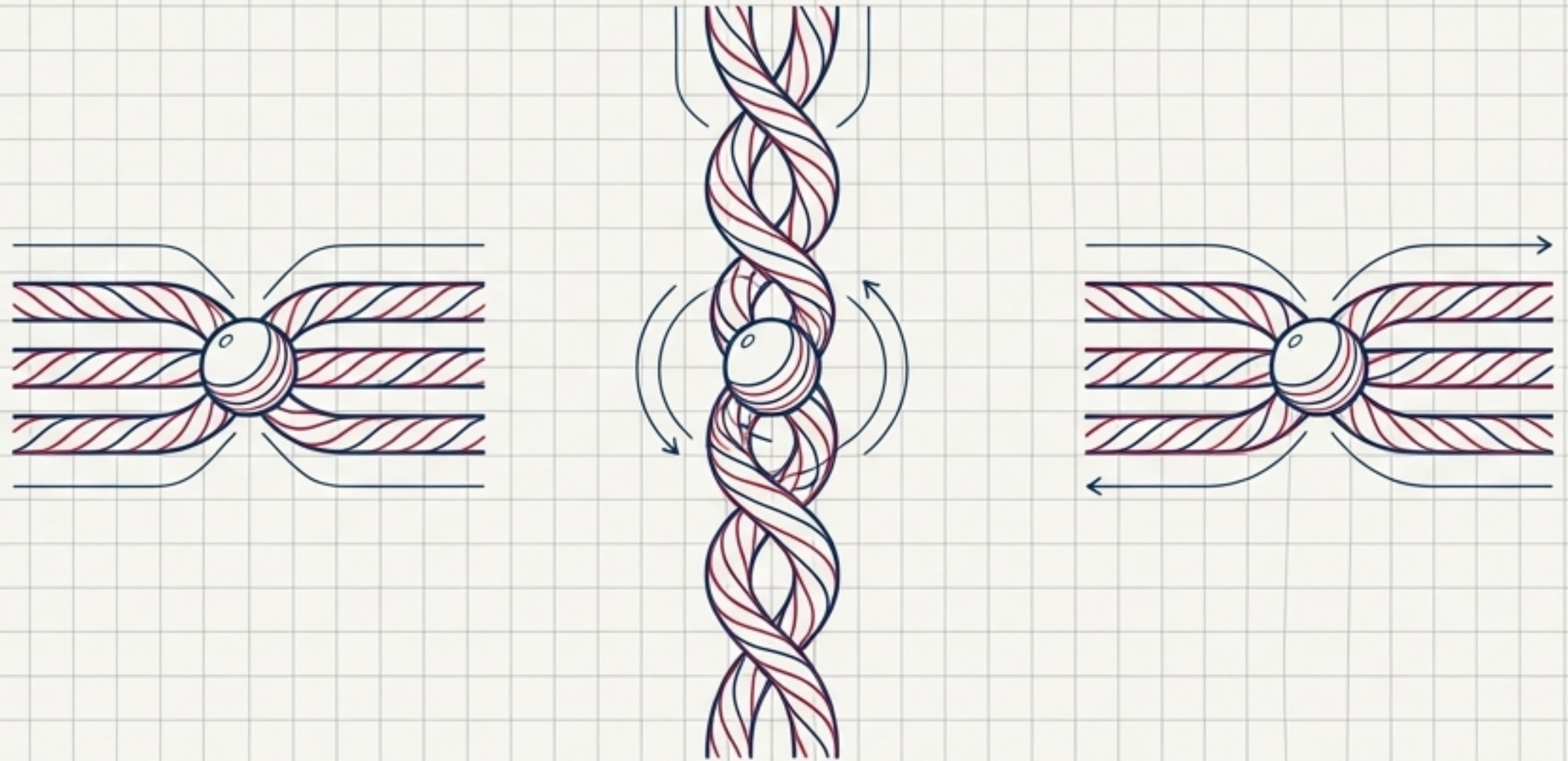
The SHD-CCP protocol models geometry as spin-1/2 fermions (quaternions).

The Macroscopic Law

Per the Dirac belt trick, a 720 degree spatial rotation is strictly required to return the entangled system to its precise, unentangled identity state.

The Directive

We explicitly bind the continuous lattice evolution to this strict 720:1 state-machine cycle.



0 deg

360 deg Rotation

720 deg (4 pi radians)

Math as a Self-Cleaning Mechanism

Synthesis Context

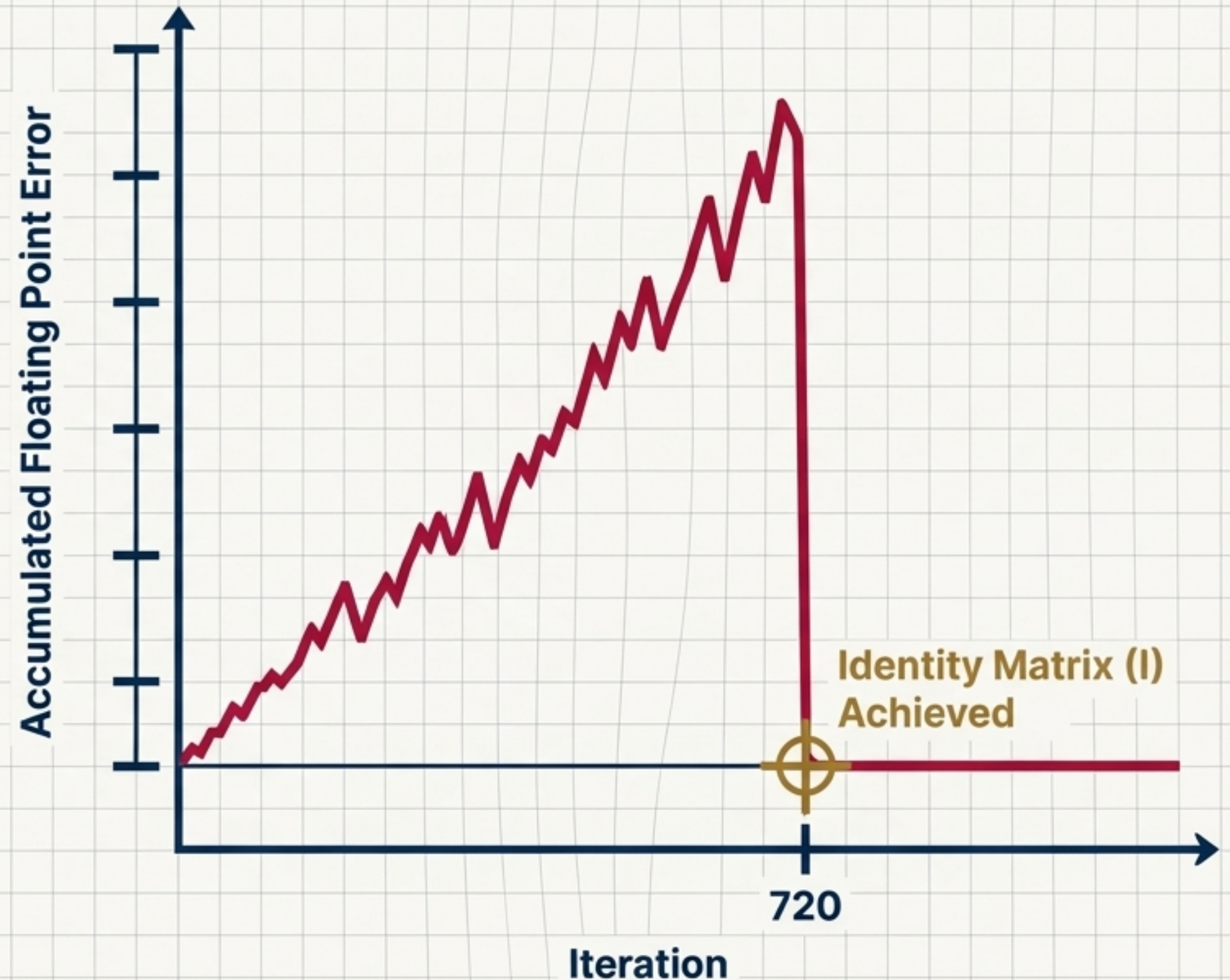
The SHD-CCP protocol is not merely simulating physics—it utilizes physics as an automated hardware zeroing function.

$$T_n = T\{n \bmod 720\}$$

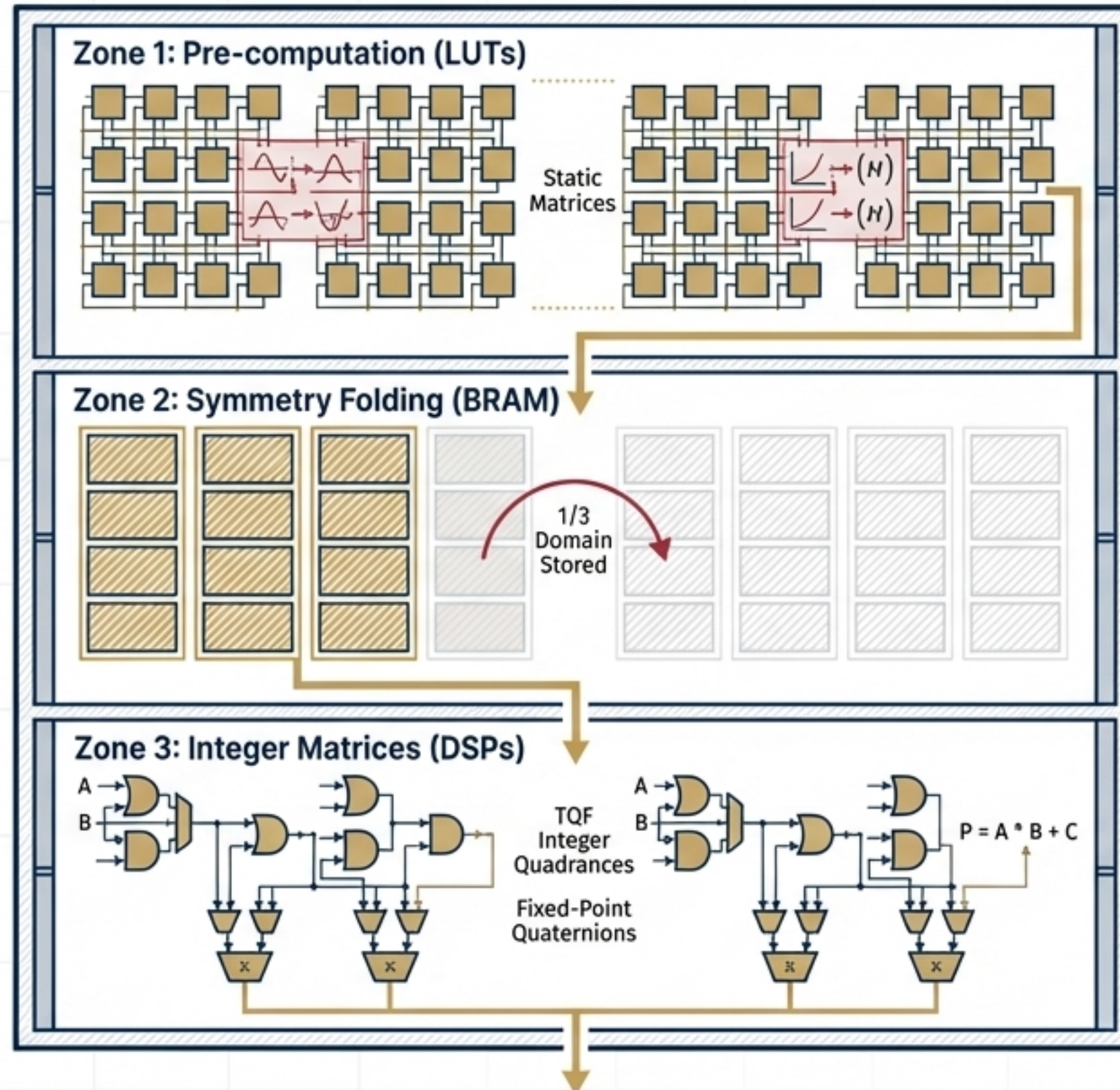
The Guarantee

At iteration 720, the exact product of all quaternion deltas, trefoil symmetries, and hyperbolic projections forces a mathematically guaranteed return to **I**.

Result: Fractional drift is violently purged without ever pausing the FPGA generation sequence.



Compiling Topology into Silicon



Zone 1: Pre-computation (LUTs)

The computationally heavy trefoil curves and hyperbolic matrices (H) are completely pre-compiled into static Look-Up Tables.

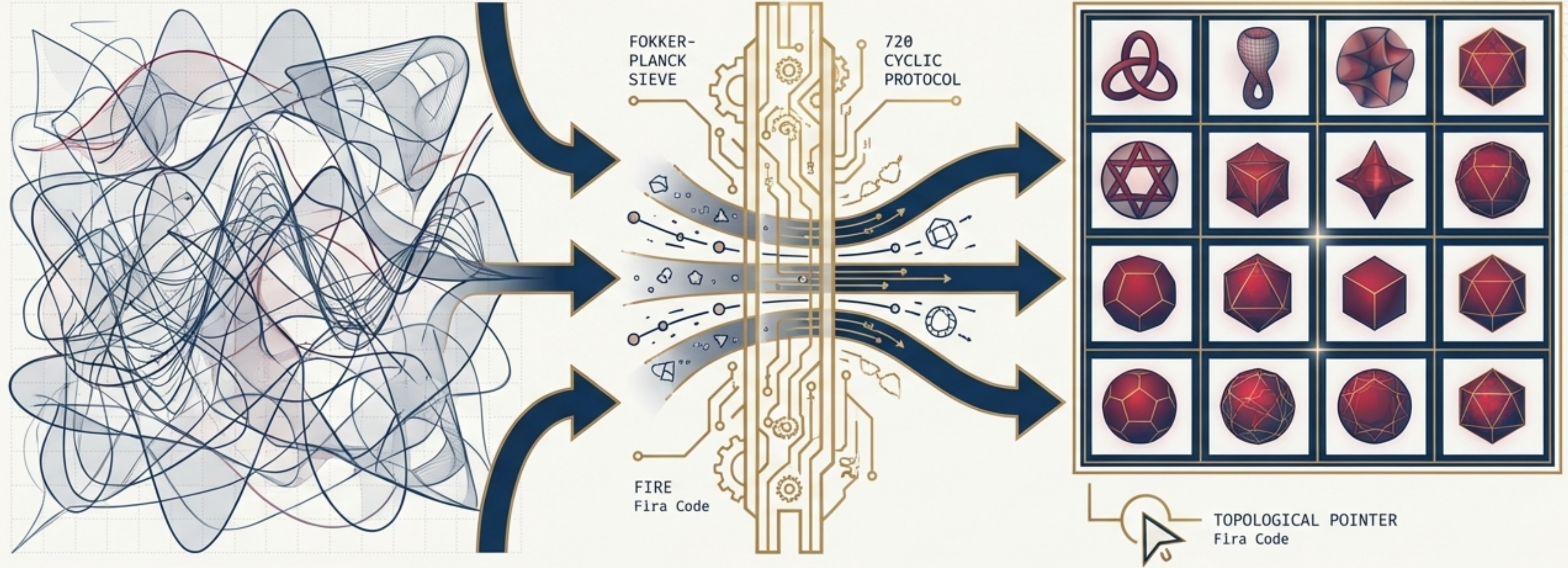
Zone 2: Symmetry Folding (BRAM)

Storing only the fundamental 1/3 domain mathematically slashes Block RAM requirements.

Zone 3: Integer Matrices (DSPs)

Native processing via DSP MAC slices. Enforcing TQF Integer Quadrances and fixed-point quaternions eliminates floating-point overhead entirely.

The Output: Constructing the Learned Codex



The Process

Once a geometric shape is targeted and stabilized by the Fokker-Planck Sieve (and defended by the 720 cyclic protocol), it is extracted.

The Codex

It is saved into the static Learned Codex as a topological pointer.

The Ultimate Paradigm

By mapping complex generation to a Codex, the system permanently replaces active calculation with static retrieval.

Future Research Horizons



The Current Triumph

The SHD-CCP protocol establishes a flawless bridge between non-Euclidean geometry and raw FPGA performance, curing computational drift permanently.

Horizon 1: Topological Invariants

Correlating knot invariants (e.g., the Jones polynomial) of the parameterized Trefoil mapping to the distribution of **prime quadrances** identified by the FP sieve.

Horizon 2: Lorentz Group Hardware Optimization

Investigating if the $SL(2, \mathbb{C})$ mapping used for hyperbolic projection can be further compressed into direct binary shift operations for exponential throughput gains.